L5

Image & Video Generation

Stable Diffusion + LoRA + 20 sample photos = infinite fashion-catalogue content. Indian e-commerce saved ₹200 crore of photography costs in 2025. Today we build that capability.

How many product photos did your last employer shoot per month?

WHAT'S INSIDE

WHY THIS LESSON MATTERS

Nykaa generates 10 lakh product images a month. Zero photoshoots.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1. Forward + reverse processes.
2. Use Stable Diffusion.
3. LoRA fine-tune on custom images.
4. SD / FLUX / DALL-E / Sora.
5. Classifier-free guidance.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01
The one-sentence intuition
Add noise, then reverse

02
Forward + reverse
Fixed vs learned

03
15-line Stable Diffusion
diffusers pipeline

04
LoRA for diffusion
Custom subject from 10 images

01

PART 1 OF 4

The one-sentence intuition

Add noise, then reverse

CONCEPT 1 OF 4

The one-sentence intuition

Add noise to an image over 1000 steps → pure static.
Train a network to reverse each step.
At inference: noise → reverse chain → image.
Hard generation problem → sequence of easy denoising problems.
TRICK
Easy denoising

Beats direct generation.

DEFINITION

The one-sentence intuition

Add noise, then reverse

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

Forward + reverse

FORWARD $q(x_t \mid x_{t-1})$ — add Gaussian noise. Fixed.
REVERSE $p(x_{t-1} \mid x_t)$ — U-Net learns to denoise.
Train: sample t; noise $x_0 \rightarrow x_t$; predict the noise.
Sample: 20-50 steps (DDIM, flow-matching schedulers).
KEY
Reverse is learned

Forward is math.

KEY POINTS

- FORWARD $q(x_t \mid x_{t-1})$ — add Gaussian noise. Fixed
- REVERSE $p(x_{t-1} \mid x_t)$ — U-Net learns to denoise
- Train: sample t; noise $x_0 \rightarrow x_t$; predict the noise
- Sample: 20-50 steps (DDIM, flow-matching schedulers)

CONCEPT 3 OF 4

15-line Stable Diffusion

StableDiffusionPipeline.from_pretrained(...)
pipe("a Banarasi saree, soft light, fashion photo")
num_inference_steps=30, guidance_scale=7.5
Save .png; share.
USE IT
15 lines

Image gen in one lecture.

DEFINITION

15-line Stable Diffusion

diffusers pipeline

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

LoRA for diffusion

DreamBooth + LoRA training.
10-20 subject images + captions.
500-1000 training steps on Colab T4.
Output: 50-200 MB LoRA file.
Plug into base pipeline; generate infinite variations.
CUSTOMISE
10 images

5 minutes of training.

KEY POINTS

- DreamBooth + LoRA training
- 10-20 subject images + captions
- 500-1000 training steps on Colab T4
- Output: 50-200 MB LoRA file

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK · PAIR · SHARE

ACTIVITY

Your first generated image

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4 Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1 *Where does your org spend most on photography?*

Q2 *What would 10,000 on-demand images unlock?*

Q3 *What ethical line would you NOT cross with generation?*

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

KNOWLEDGE CHECK · QUESTION 1 OF 3

Forward process:

A Learned

B Adds noise; fixed

C Denoises

D Random

ANSWER · QUESTION 1 OF 3

Forward process:

B

CORRECT ANSWER

Adds noise; fixed

WHY

Forward is mathematical noising; reverse is learned

KNOWLEDGE CHECK · QUESTION 2 OF 3

Typical 2026 sampling steps:

A B

B D

C Answer: B. DDIM/Euler/flow-matching cut 1000 to 20-50.

D (no option)

ANSWER · QUESTION 2 OF 3

Typical 2026 sampling steps:

CORRECT ANSWER

B

D

WHY

The correct option matches the lesson's core principle.

KNOWLEDGE CHECK · QUESTION 3 OF 3

LoRA for diffusion size:

- **A** 1 KB
- **B** 1 MB
- **C** 50-200 MB
- **D** 10 GB

ANSWER · QUESTION 3 OF 3

LoRA for diffusion size:

CORRECT ANSWER

C

50-200 MB

WHY

Small enough to ship; large enough to capture identity

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

Add noise to an image over 1000 steps → pure static.
Train a network to reverse each step.
At inference: noise → reverse chain → image.
Hard generation problem → sequence of easy denoising problems.
TRICK
Easy denoising

Beats direct generation.

THE TAKEAWAY

"Everything else is tuning."

WHAT TO NOTICE

1. The one-sentence intuition
2. Forward + reverse
3. 15-line Stable Diffusion
4. LoRA for diffusion

APPLY IT → ASSIGNMENT

Fine-tune on Banarasi sarees

15 sample photos.

DELIVERABLES

- Collect + caption.
- LoRA train 500 steps.
- Generate 5 lifestyle images.
- Post.

WHAT 'GOOD' LOOKS LIKE

Deliverable: 5 images on forum.

ONE THING TO REMEMBER

“

"Everything else is tuning."

— AI Learning Hub

END OF LESSON 7

What you can now do

Each one of these is now part of your working vocabulary.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

NEXT → Lesson 6 — Edge AI.